

# Performance Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators

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## Abstract

This paper presents a measurement-only performance benchmark for embedded battery state estimation under realistic Battery Management System (BMS) constraints. The study isolates estimator performance rather than hardware-resource or implementation-effort evaluation and compares four embedded-ready estimators on the same Lithium Iron Phosphate (LFP) full-cell trajectory under a shared embedded constraint: a direct-measurement model (DM), a hybrid direct-measurement model with learned State-of-Health support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with GRU-based SOC estimation and LSTM-based online SOH context (DD). The benchmark perturbs only measurable inputs and timing information, including ADC quantization, current bias, additive current, voltage, and temperature noise, missing samples, irregular sampling, burst dropout, voltage spikes, and initial-state mismatch, so that all disturbances remain physically interpretable and reproducible. The comparison reveals no universal winner across deployment-relevant dimensions. Under clean conditions, HDM provides the lowest baseline error with an MAE of 0.0024, followed by HECM with 0.0090, DD with 0.0100, and DM with 0.0311. Under disturbed measurements, HECM is the strongest overall robustness choice, especially in the most discriminative current-bias and voltage-noise scenarios. DD is the most stable estimator under missing samples and recovers fastest after initialization mismatch, while DM

shows the expected drift sensitivity and HDM trades nominal accuracy against reduced disturbance tolerance. The benchmark therefore exposes complementary failure modes of direct-measurement, hybrid physics-based, and data-driven estimators and supports decision-oriented model selection for embedded BMS deployment.

*Keywords:* Battery management system, State of charge, State of health, Performance benchmark, Direct measurement, Equivalent circuit model, Data-driven estimation, Embedded artificial intelligence

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## 1 Nomenclature

### 2 *Acronyms*

|    |             |                                 |
|----|-------------|---------------------------------|
| 3  | <b>BMS</b>  | Battery management system       |
| 4  | <b>SOC</b>  | State of charge                 |
| 5  | <b>SOH</b>  | State of health                 |
| 6  | <b>DM</b>   | Direct-measurement model        |
| 7  | <b>HDM</b>  | Hybrid direct-measurement model |
| 8  | <b>HECM</b> | Hybrid equivalent-circuit model |
| 9  | <b>DD</b>   | Data-driven neural model        |
| 10 | <b>ECM</b>  | Equivalent circuit model        |
| 11 | <b>EKF</b>  | Extended Kalman filter          |
| 12 | <b>MAE</b>  | Mean absolute error             |
| 13 | <b>RMSE</b> | Root-mean-square error          |
| 14 | <b>ADC</b>  | Analog-to-digital converter     |
| 15 | <b>CV</b>   | Constant-voltage phase          |
| 16 | <b>OCV</b>  | Open-circuit voltage            |
| 17 | <b>GRU</b>  | Gated recurrent unit            |
| 18 | <b>LSTM</b> | Long short-term memory          |
| 19 | <b>LFP</b>  | Lithium iron phosphate          |

## 20 *Symbols*

|    |                              |                                                         |
|----|------------------------------|---------------------------------------------------------|
| 21 | $t$                          | Time                                                    |
| 22 | $I(t)$                       | Current signal at time $t$                              |
| 23 | $U(t)$                       | Voltage signal at time $t$                              |
| 24 | $T(t)$                       | Temperature signal at time $t$                          |
| 25 | $\widehat{\text{SOC}}(t)$    | Estimated state of charge at time $t$                   |
| 26 | $\widehat{\text{SOH}}(t)$    | Estimated state of health at time $t$                   |
| 27 | $\text{SOC}_{\text{ref}}(t)$ | Reference SOC target at time $t$                        |
| 28 | $\text{SOH}_{\text{ref}}(t)$ | Reference SOH target at time $t$                        |
| 29 | $\Delta I$                   | Current bias or offset                                  |
| 30 | $\Delta U$                   | Voltage bias or offset                                  |
| 31 | $\Delta T$                   | Temperature bias or offset                              |
| 32 | $\sigma_I$                   | Standard deviation of injected current noise            |
| 33 | $\sigma_U$                   | Standard deviation of injected voltage noise            |
| 34 | $\sigma_T$                   | Standard deviation of injected temperature noise        |
| 35 | $Q_c$                        | Online cumulative charge state used by the SOC models   |
| 36 | $C_{\text{nom}}$             | Nominal capacity                                        |
| 37 | $C_{\text{eff}}$             | Effective capacity used inside the estimator            |
| 38 | $U_{\text{max}}$             | Upper voltage threshold used for full-charge detection  |
| 39 | $U_{\text{tol}}$             | Voltage tolerance used in the full-charge condition     |
| 40 | $U_1, U_2$                   | Polarization voltages of the first and second RC branch |
| 41 | $\Delta\text{MAE}$           | Error increase relative to baseline                     |

## 42 **1. Introduction**

### 43 *1.1. Motivation and practical relevance*

44     Reliable battery state estimation is a core BMS function because SOC and  
45     SOH determine safety margins, usable energy, operating limits, charge con-  
46     trol, and degradation-aware system management in both mobile and stationary

lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not defined by clean-condition accuracy alone, because real BMS operation is exposed to sensor bias, additive noise, initialization uncertainty after restart, missing samples, irregular sampling, communication interruptions, and transient signal spikes caused by measurement-chain limitations or disturbances in the surrounding system [4, 5, 6]. The relevant engineering question is therefore not only whether an estimator achieves low MAE under nominal measurements, but also whether it converges back to a reliable state after disturbed initialization and whether it remains stable under physically plausible measurement corruption.

As summarized conceptually in Figure 1, the present benchmark treats deployment-facing performance as a three-dimensional problem consisting of nominal accuracy, convergence behavior after state inconsistency, and robustness against disturbed measurements and signal-integrity faults, embedded within broader BMS requirements such as hardware constraints and implementation effort. The present study therefore isolates the performance branch and compares representative estimator classes under identical measurement-side disturbances and a matched embedded-feasibility constraint. The benchmark shows that strong nominal accuracy does not necessarily imply strong disturbance performance and that the estimator classes respond differently to bias, missing samples, and re-initialization events.

## 1.2. State of the art in battery state estimation

The literature commonly groups battery-state estimators into direct-measurement or integration-based approaches, model-based approaches based on equivalent-circuit or electrochemical models, hybrid approaches that combine physical structure with learned components, and purely data-driven models based on neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically based on Coulomb counting, are attractive because they are simple, computationally inexpensive, and easy to deploy; however, they remain structurally sensitive to current bias, missing data, and initialization errors because they lack an internal correction mechanism [7, 4]. Model-based approaches, especially ECM-

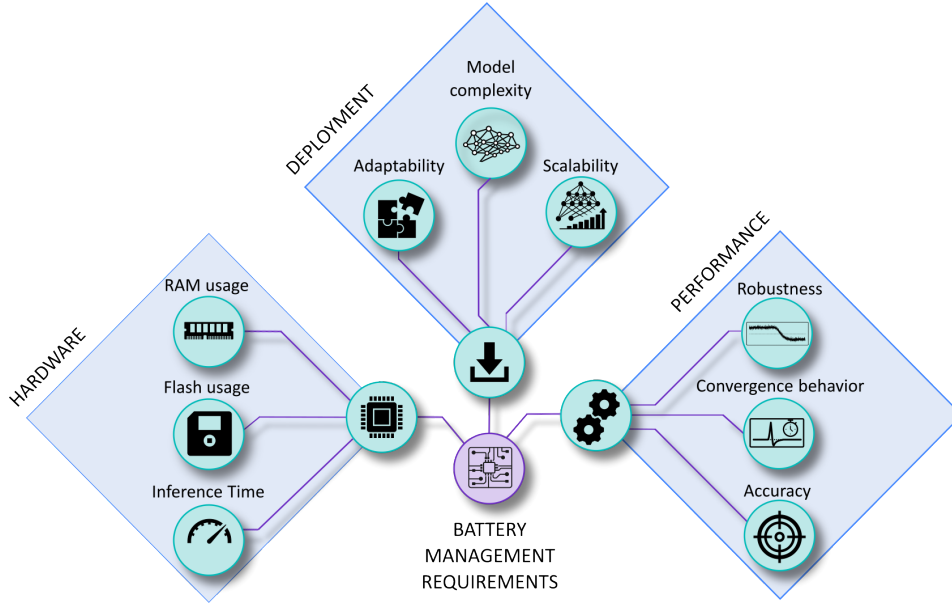


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

77 and observer-based estimators, offer physical interpretability and voltage-based  
78 correction capability, but their practical behavior depends strongly on the qual-  
79 ity of model parametrization, operating-point observability, and measurement  
80 quality [8, 2, 4].

81 Hybrid estimators seek to combine physical state propagation with learned  
82 adaptation of capacity, parameters, or auxiliary states, which has made joint  
83 SOC/SOH estimation an active research direction [9, 10, 11]. This combination  
84 can improve baseline calibration, while it does not necessarily eliminate struc-  
85 tural sensitivity to measurement disturbances. Data-driven approaches based  
86 on recurrent or sequence models can exploit nonlinear cross-relations between  
87 current, voltage, temperature, and derived online features, and recent work has  
88 also demonstrated their embedded feasibility through TinyML-oriented imple-

89 mentations, pruning, and quantization [12, 13, 14, 15, 3].

### 90 *1.3. Performance and embedded deployment gap*

91 A large share of the SOC estimation literature still emphasizes clean-data ac-  
92 curacy, average error metrics, or computational efficiency, while comparatively  
93 fewer studies analyze how estimators behave under realistic measurement faults  
94 and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain  
95 confined to a single model family, for example observer-based robustness against  
96 modeling and measurement errors [4] or robust estimation concepts for specific  
97 BMS architectures under noise and uncertainty [5], whereas cross-family com-  
98 parisons under the same disturbance protocol and under comparable embedded  
99 constraints remain limited. Benchmarking studies in the broader BMS con-  
100 text underline the importance of reproducible scenario design and deployment-  
101 relevant validation, but they do not by themselves close the gap between clean-  
102 data estimator comparison and measurement-level performance benchmarking  
103 across fundamentally different estimator classes [6]. This gap is especially rele-  
104 vant in embedded BMS design, where estimator selection must jointly consider  
105 nominal accuracy, convergence, robustness under disturbed measurements, and  
106 computational constraints.

### 107 *1.4. Contribution and study objectives*

108 This work defines deployment-facing estimator performance as the joint as-  
109 sessment of nominal accuracy, convergence behavior after state inconsistency,  
110 and robustness under realistic disturbances applied only to measurable inputs  
111 and their timing. The study compares four deployment-prepared representa-  
112 tives of major estimator families, namely a direct-measurement model (DM), a  
113 hybrid direct-measurement model with learned SOH support (HDM), a hybrid  
114 equivalent-circuit model with voltage feedback (HECM), and a data-driven neu-  
115 ral model with rolling-window GRU-based SOC estimation and stateful LSTM-  
116 based online SOH context (DD), under identical causal execution conditions.

117 The disturbances are deliberately restricted to measurement-level and signal-  
 118 integrity scenarios, including current bias, current noise, voltage noise, temper-  
 119 ature noise, missing samples, irregular sampling, burst dropouts, voltage spikes,  
 120 ADC quantization, and initialization errors, so that the benchmark remains  
 121 physically interpretable and experimentally reproducible.

122 The manuscript should therefore be read as a controlled cross-class esti-  
 123 mator benchmark and not as a complete system-level BMS validation cam-  
 124 paign. Its role is to isolate estimator-side failure modes under matched data,  
 125 matched disturbance definitions, matched online execution rules, and matched  
 126 embedded-feasibility constraints. The resulting contribution is a measurement-  
 127 only benchmark that separates baseline accuracy, convergence, and disturbed-  
 128 operation robustness and translates the observed failure modes into decision-  
 129 oriented guidance for embedded BMS estimator selection.

## 130 **2. Methodology**

### 131 *2.1. Estimator classes under test*

132 Figure 2 summarizes the causal benchmark chain, from measured input sig-  
 133 nals and online feature reconstruction to disturbance injection and the final  
 134 global and local performance evaluation. The direct-measurement model (DM)  
 135 represents the canonical embedded Coulomb-counting approach with fixed nom-  
 136 inal capacity and explicit full-charge reset logic, and serves as the low-complexity  
 137 reference for integration-driven SOC estimation. All SOH-dependent estima-  
 138 tor variants in the benchmark, namely HDM, HECM, and DD, use the same  
 139 shared LSTM-based online SOH estimator; the classes differ only in how this  
 140 common SOH signal is consumed inside the SOC pipeline. The hybrid direct-  
 141 measurement model (HDM) preserves the same Coulomb-counting core but re-  
 142 places the fixed capacity assumption with an online effective capacity derived  
 143 from this shared SOH estimator, which makes it representative of lightweight  
 144 hybrid estimation. The hybrid equivalent-circuit model (HECM) represents  
 145 physics-based state estimation with observer correction, because it combines

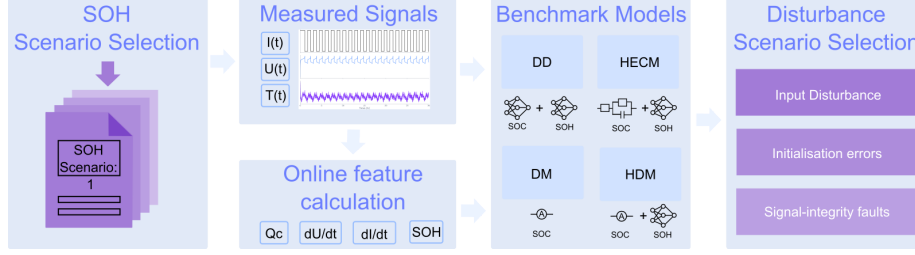


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator classes, exposed to measurement-only disturbances, and finally evaluated through global and local performance metrics.

146 current-driven state propagation with voltage-residual feedback and SOH-dependent  
 147 parameterization in a second-order RC equivalent-circuit structure driven by the  
 148 same shared SOH estimate [8, 2]. The data-driven model (DD) represents a neu-  
 149 ral estimator pipeline in which SOC is predicted by a lightweight rolling-window  
 150 GRU sequence model while the shared LSTM-based online SOH estimate is pro-  
 151 vided as causal context under the same execution constraints [12, 3].

## 152 2.2. SOC estimation methods

### 153 2.2.1. Direct-measurement model (DM)

The direct-measurement SOC formulation used in DM, HDM, and as an explicit auxiliary feature in DD follows the same online cumulative-charge state, denoted here uniformly as  $Q_c$ . It is updated causally from the measured current stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

where  $I_k$  is given in ampere and  $\Delta t_k$  in seconds, so that the factor  $3600 \text{ s h}^{-1}$  converts the increment to ampere-hours. The sign convention is fixed by the benchmark implementation and the state is reset to  $Q_c = 0$  after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right), \quad (2)$$



154 *2.2.2. Hybrid direct-measurement model (HDM)*

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

155 This formulation makes the relation between DM and HDM explicit: both use  
 156 the same online charge state  $Q_c$ , while only the denominator changes through  
 157 online SOH support. In both direct-measurement variants, a sustained near-  
 158 maximum voltage condition is interpreted as a constant-voltage full-charge event;  
 159 if the voltage remains above  $U_{\text{max}} - U_{\text{tol}}$ , where  $U_{\text{tol}}$  denotes the admissible volt-  
 160 age margin below the upper threshold  $U_{\text{max}}$ , for the configured CV duration, the  
 161 internal charge state is reset so that  $Q_c = 0$  and therefore  $\widehat{\text{SOC}} = 1$ , which re-  
 162 flects the practical assumption that a completed CV phase provides a physically  
 163 anchored full-charge reference.

164 *2.2.3. Hybrid equivalent-circuit model (HECM)*

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \quad (5)$$

where  $U_1$  and  $U_2$  denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6)$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta \Delta t_k / C_b \\ R_1 (1 - e^{-\Delta t_k/\tau_1}) \\ R_2 (1 - e^{-\Delta t_k/\tau_2}) \end{bmatrix}, \quad (7)$$

where  $C_b = C_0 \widehat{\text{SOH}}$  is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\widehat{U}_k = \text{OCV}(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \quad (8)$$

and the Kalman update uses the residual between  $\widehat{U}_k$  and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample  $I_{k-1}$ , whereas the measurement equation uses the current sample  $I_k$  available at the correction step. The ECM parameters are not constant. The implementation interpolates  $R_i$ ,  $R_1$ ,  $R_2$ ,  $\tau_1$ ,  $\tau_2$ , OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

#### 2.2.4. Data-driven SOC model (DD)

The data-driven SOC model is a GRU-based recurrent estimator with the feature vector  $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{SOH}, Q_c, dU/dt, dI/dt, \Delta t\}$ . The benchmarked structurally pruned and fine-tuned checkpoint consists of a single-layer GRU with hidden size 67 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length  $L_{\text{SOC}} = 2024$  samples. Within this feature set,  $Q_c$  is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as finite differences with respect to the current sampling interval,

$$\frac{dI}{dt}(k) = \frac{I_k - I_{k-1}}{\Delta t_k}, \quad \frac{dU}{dt}(k) = \frac{U_k - U_{k-1}}{\Delta t_k}, \quad (9)$$

and provide the neural estimator with local transient information on current and voltage dynamics that is not directly captured by the absolute channel values alone. The additional feature  $\Delta t_k$  is passed explicitly so that the network can

distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. No future information is used. Continuous hidden-state forwarding is evaluated only as a separate deployment ablation because it changes the learned inference semantics.

### 2.3. SOH estimation method

The shared SOH estimator used in the current benchmark is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from  $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{EFC}, Q_c\}$ , where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information; the first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and then held piecewise constant between update points, which enforces an online deployment constraint and avoids sample-wise access to future information while still allowing the faster SOC estimators to use the latest SOH context. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation  $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$ , so the method remains structurally close to direct measurement while replacing

Table 1: Estimated persistent flash and RAM footprints of the reusable estimator components for the deployment-prepared variants. For the recurrent neural components, flash estimates refer to the structured-pruned, finetuned, and 8-bit quantized models. RAM\* includes only architecture-intrinsic persistent numerical states; temporary buffers, framework overhead, stack usage, and system-level runtime memory are excluded.

| Component                  | Flash [KiB] | RAM* [B] | Basis                                                                                                   |
|----------------------------|-------------|----------|---------------------------------------------------------------------------------------------------------|
| DM core (Coulomb counting) | 4.2         | 16       | four persistent scalar states in the online update logic                                                |
| SOC-GRU core               | 27.9        | 268      | one recurrent hidden state with size 67 stored as float32                                               |
| SOH-LSTM core              | 408.3       | 1792     | two-layer LSTM with hidden and cell states of size 112 stored as float32                                |
| HECM core (2RC observer)   | 442.5       | 52       | observer state $x \in \mathbb{R}^3$ , covariance $P \in \mathbb{R}^{3 \times 3}$ , and previous current |

the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and the lookup-based ECM parametrization, while in the DD model it is provided as an explicit SOC input feature that contextualizes the current measurement trajectory.

#### 2.4. Embedded-oriented model preparation

The focus of the present paper is estimator performance and not hardware-oriented model optimization. The compared neural checkpoints use compact widths selected before the benchmark: hidden size 67 for the structurally pruned and fine-tuned SOC-GRU and hidden size 112 for the shared SOH-LSTM. The primary software benchmark evaluates these floating-point checkpoints; it does not substitute a separately quantized checkpoint. Tables 1 and 2 therefore provide analytical parameter/state-storage estimates rather than measured proof of target deployment. Actual latency, peak runtime RAM including buffers and stack, flash image size, and CPU load remain part of the separate STM32 hardware campaign.

Table 2: Estimated combined estimator footprints relative to a reference embedded budget of 1 MB flash and 1 MB RAM. The recurrent neural components correspond to the deployment-prepared variants after structured pruning, finetuning, and 8-bit quantization. RAM\* reports only persistent numerical states and therefore serves as a strict lower bound rather than as a full runtime-memory measurement.

| Estimator | Flash [KiB] | RAM* [B] | Flash < 1 MB | RAM* < 1 MB |
|-----------|-------------|----------|--------------|-------------|
| DM        | 4.2         | 16       | yes          | yes         |
| HDM       | 412.5       | 1808     | yes          | yes         |
| HECM      | 850.8       | 1844     | yes          | yes         |
| DD        | 436.2       | 2060     | yes          | yes         |

## 2.5. Performance benchmark design

The benchmark follows a strict measurement-only principle: only quantities that a real BMS measures or derives online from measured data are manipulated, namely current, voltage, temperature, sample availability, and sampling time, whereas artificial parameter mismatch and synthetic hidden-state corruption are deliberately excluded [6]. Figure 3 groups the benchmark into input disturbances, initialization errors, and signal-integrity faults, while Table 3 defines the exact disturbance implementations used in the study.

The disturbance magnitudes are not intended as one-to-one copies of a single field specification. Instead, they were selected from three justification classes: deployment-level sensing limitations, adverse measurement corruption, and deliberate signal-integrity stress tests [20, 21, 22, 23, 24, 25]. The main benchmark matrix follows the scenario families shown in Figure 3 and provides the basis of the main result figures and tables. ADC quantization is part of the same scenario set and is included in the cross-scenario overview; its detailed numerical results are reported in Appendix Table B.9 and Figure B.20. Overall, the benchmark is designed as a controlled and reproducible scenario study that exposes estimator-specific failure modes under a common disturbance protocol rather than emulating the complete system stack of a production BMS.

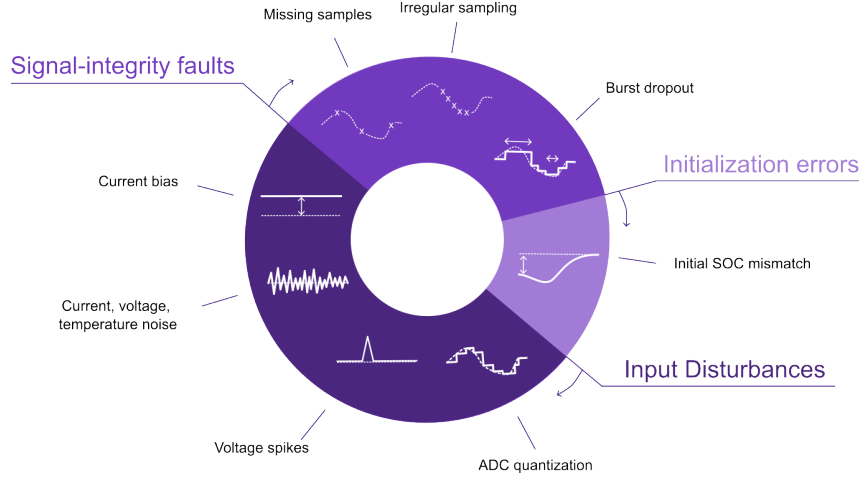


Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

Table 3: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings; the cited sources motivate disturbance type and magnitude range.

| Scenario           | Benchmark setup                                                                                             | Practical origin / rationale                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| ADC quantization   | $\Delta I = 0.01 \text{ A}$ ; $\Delta U = 0.005 \text{ V}$ ; $\Delta T = 0.5 \text{ }^\circ\text{C}$        | ADC steps; sensor scaling; rounding; coarse deployment-level discretization [20, 21, 26] |
| Current gain error | Signed gain sweep: $I_{\text{meas}} = I(1 + b)$ with $b = \pm 0.5\%$ , $\pm 1.5\%$ , and $\pm 3.0\%$        | Sensor gain error; miscalibration; drift; upper bound as stress case [21, 22, 23]        |
| Current noise      | Gaussian; $\sigma_I = 0.02 \text{ A}$ and $0.10 \text{ A}$                                                  | Sensing-chain noise; EMI pickup; low and high sensor-noise levels [4, 5]                 |
| Voltage noise      | Gaussian; $\sigma_U = 0.01 \text{ V}$                                                                       | Wiring noise; front-end noise; poor filtering; EMI [4, 20, 24]                           |
| Temperature noise  | Gaussian; $\sigma_T = 1.0 \text{ }^\circ\text{C}$                                                           | Sensor tolerance; conversion noise; thermal gradients; sensor-to-cell mismatch [26, 11]  |
| Initial SOC error  | Initial mismatch: $-0.10 \text{ SOC}$ ; DD via perturbed online $Q_c$                                       | Restart; storage; missing recent history; weak LFP OCV anchoring [4, 8]                  |
| Missing samples    | Every 50th sample frozen; separate random 2% loss                                                           | Packet loss; blocked updates; logger dropouts; periodic and random data freeze [5, 25]   |
| Irregular sampling | Uniform jitter: $\pm 0.1 \text{ s}$ , $\pm 0.5 \text{ s}$ , and $\pm 0.9 \text{ s}$ around nominal 1 s grid | Scheduling jitter; bus latency; asynchronous acquisition [6, 5, 25]                      |

Table 3: Disturbance configurations used in the benchmark and practical rationale (continued).

| Scenario       | Benchmark setup                                                | Practical origin / rationale                                                            |
|----------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Burst dropout  | One central frozen gap: 3600 s                                 | Communication interruption; frozen task; stuck data stream; severe recovery case [6, 5] |
| Voltage spikes | Single-sample spikes: $\pm 0.20$ V every 1000 samples (1000 s) | EMI; switching disturbance; corrupted voltage samples [24]                              |

## 2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, convergence behavior after initialization mismatch or interrupted signal continuity, and robustness under sustained or transient measurement corruption. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error, because these metrics provide a compact view of average performance, systematic deviation, and tail behavior across the selected evaluation windows. Local disturbance performance is analyzed through scenario-specific metrics such as jump counts, output variance, absolute-error variance, and excess rolling MAE, because several disturbances do not primarily manifest as window-wide MAE collapse but as short-term output fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance mask exists, the analysis further separates pre-disturbance, disturbed, and post-disturbance behavior in order to quantify disturbance-window error growth, recovery time, and residual post-disturbance error.

ADC quantization, current bias, voltage noise, temperature noise, missing samples, and irregular sampling are interpreted primarily through global metrics, whereas burst dropout, voltage spikes, current-noise detail, and initialization mismatch additionally use dedicated local evidence because their most relevant effects are episodic or recovery-related. Recovery comparisons use one estimator-independent criterion: absolute SOC error below 0.02 for a continuous 300 s, evaluated up to a common 24 h censoring horizon. Runs that do

not recover are retained as censored observations rather than silently omitted. The comparison reports recovery-or-censoring time, the censored fraction, initial post-event error, MAE after 1 h and 6 h, and excess-error area under the curve. The requested  $-0.10$  initialization mismatch is mapped into each estimator’s available online state: the coulomb-counting SOC initialization for DM and HDM, the EKF SOC state for HECM, and the capacity-equivalent  $Q_c$  offset for DD. Because the DD mapping passes through the recurrent network rather than directly fixing its output, the realized initial output error is reported for every model before recovery is compared.

The independent statistical unit is the holdout cell, not an individual time sample or selected window. Predeclared SOH windows are first averaged within each cell and random seed; the six cells then receive equal macro weight. Hierarchical bootstrap resampling with seeds nested inside cells provides 95% confidence intervals. Scenario effects are evaluated as paired, cell-matched differences from baseline, and estimator pairs are compared using exact two-sided sign-flip tests, raw mean and median differences, paired standardized effect sizes, and Holm-adjusted  $p$ -values. Figure 4 summarizes this analysis chain.

### 3. Experimental Setup

#### 3.1. Dataset and data acquisition

The benchmark is based on the MGFarm LFP 18650 dataset, which was generated in a laboratory environment as a controlled design-of-experiments study on cyclic aging and operating conditions and combines long full-cell trajectories with high-resolution measured signals and repeated checkup structures [27, 28, 29]. Model development and final testing use a fixed cell-disjoint split. SOC and SOH training use C01, C03, C05, C11, C17, and C23; validation uses C07, C19, and C21; and the independent holdout benchmark uses C09, C13, C15, C25, C27, and C29. The six test cells span distinct aging, thermal, duration, and load envelopes, as summarized in Figure 5. Conclusions are therefore



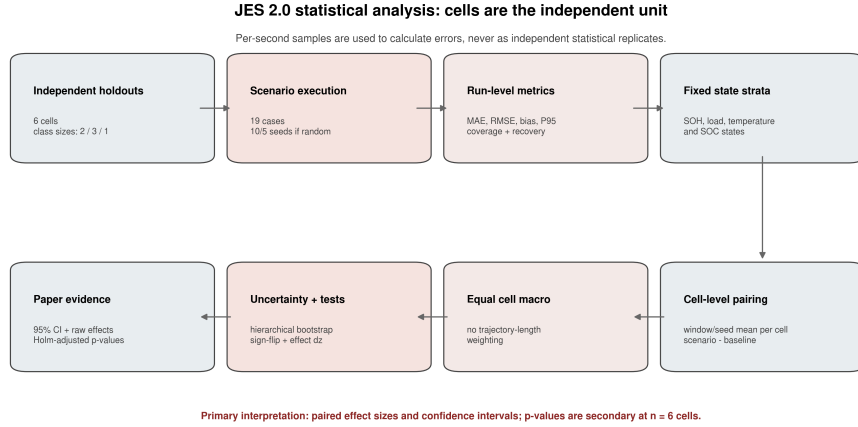


Figure 4: Statistical analysis workflow used for JES2. Measurement samples form the windows from which run-level errors are calculated, whereas cells remain the independent units for uncertainty estimation and paired inference. Windows are averaged within cell and seed, random seeds are nested within cells, and final reporting emphasizes paired effects and confidence intervals.

limited to the available LFP chemistry and measured operating envelope; the study does not claim cross-chemistry or pack-level generalization.

The holdout cells are assigned to descriptive load groups before model results are inspected, using visible separation in the measured 95th-percentile absolute  $C$ -rate: C25 and C27 form the low-load group, C09, C13, and C15 the middle-load group, and C29 the high-load case. Because the high-load group contains one cell, its group-by-SOH results are explicitly exploratory and are not interpreted as population-level high-load inference. Within each trajectory, performance is additionally stratified by reference SOH (fresh:  $\text{SOH} \geq 0.90$ ; mid-life:  $0.80 \leq \text{SOH} < 0.90$ ; aged:  $\text{SOH} < 0.80$ ), instantaneous absolute  $C$ -rate ( $< 0.5$ ,  $0.5\text{--}1.5$ , and  $\geq 1.5$ ), temperature ( $\leq 30^\circ\text{C}$ ,  $30\text{--}35^\circ\text{C}$ , and  $> 35^\circ\text{C}$ ), and SOC ( $< 0.20$ ,  $0.20\text{--}0.80$ , and  $> 0.80$ ). Figures 6–8 make the resulting coverage and missing combinations explicit; absent states are reported as not available and are never imputed.

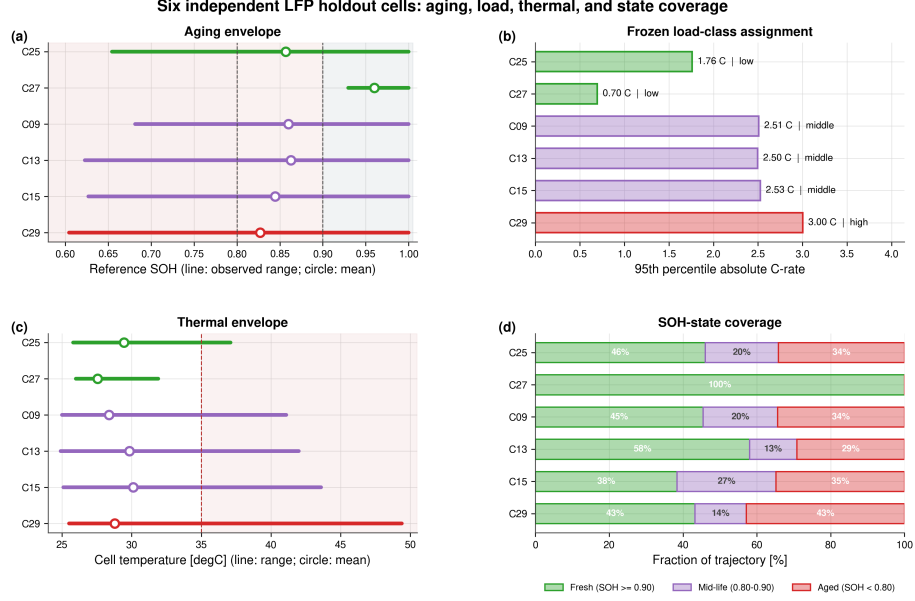


Figure 5: Quantitative coverage of the six independent holdout cells. Panel (a) shows the observed SOH range and mean, panel (b) the frozen load-class assignment based on the measured 95th-percentile absolute C-rate, panel (c) the measured temperature range and mean, and panel (d) the fraction of each full trajectory in the predeclared fresh, mid-life, and aged SOH states. The load classes are descriptive strata fixed independently of estimator errors.

### 3.2. Reference targets and preprocessing

The reference preprocessing first computes the signed transferred charge as  $Q_k = I_k \Delta t_k / 3600$  and the absolute transferred charge as  $|Q_k|$ , from which the capacity during dedicated capacity-test intervals is obtained as  $C_{\text{ref}} = \sum_k |Q_k|$ . The reference SOH target is then defined offline as  $\text{SOH}_{\text{ref}} = C_{\text{ref}} / C_{\text{ref},0}$ , where  $C_{\text{ref},0}$  denotes the first measured reference capacity of the trajectory, and the resulting capacity and SOH columns are interpolated between the dedicated capacity-test anchors. The reference SOC target used in the benchmark corresponds to the offline Coulomb-counting target  $\text{SOC}_{\text{ref}} = 1 + Q_c / C_{\text{ref},0}$ , where the cumulative charge state  $Q_c$  is propagated from the signed charge and reset to 0 at the upper voltage anchor and to  $-C_{\text{ref},0}$  at the lower voltage anchor dur-

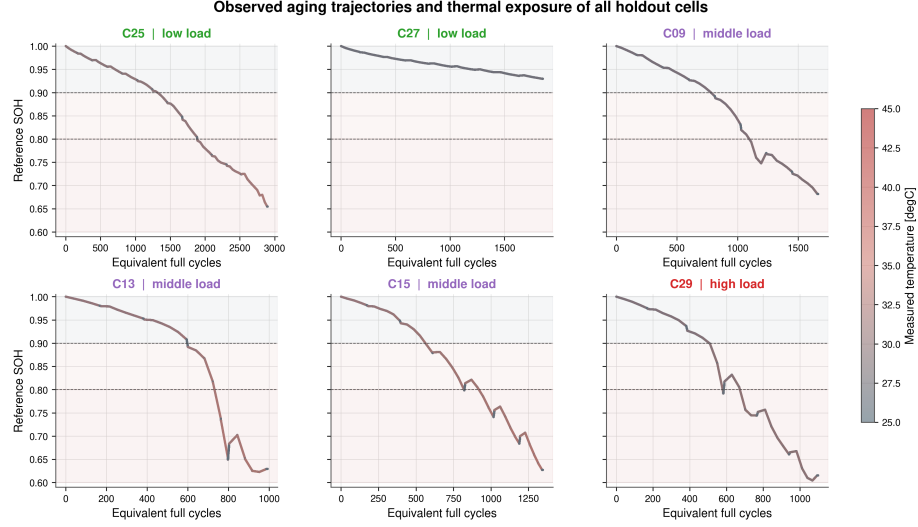


Figure 6: Observed reference-SOH trajectories over equivalent full cycles for all six holdout cells. Point color represents measured temperature. C27 provides a mild-aging control, whereas the remaining cells cover mid-life and aged operation with different degradation trajectories.

ing preprocessing. These reference targets are intentionally treated as offline evaluation quantities, because they rely on dataset-side processing and anchor logic that are not fully available to an online estimator during deployment.

### 3.3. Test trajectory and benchmark scenarios

All estimators are evaluated on every available holdout-cell/SOH combination under the same disturbance protocol. One representative 24-h window is frozen for each available fresh, mid-life, and aged state, yielding 16 windows; C27 contributes only a fresh window because its measured SOH does not fall below 0.90. Windows start at a measured full-charge anchor ( $\text{SOC} \geq 0.98$ ,  $U \geq 3.58 \text{ V}$ ) and are selected as measured-feature medoids without using estimator outputs or errors. The one-hour dropout uses 48 h from the same anchor to retain approximately 24 h of post-gap observation. The shared SOH LSTM receives 192 h of undisturbed causal context before every scored window, matching its trained sequence length. Figure 9 summarizes this protocol.

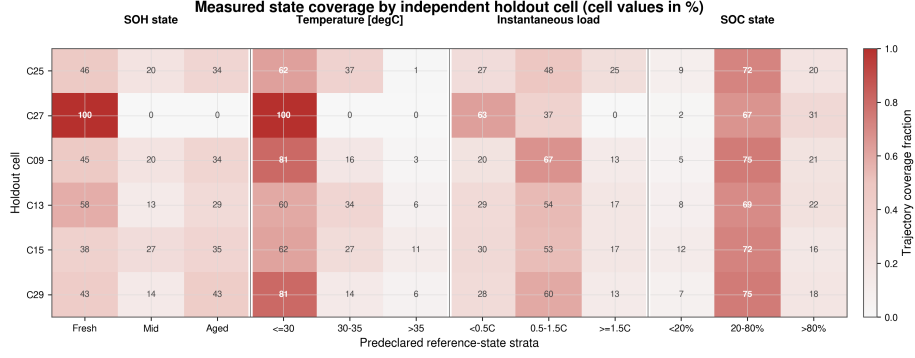


Figure 7: Fraction of each complete holdout trajectory in the predeclared SOH, temperature, instantaneous-load, and SOC strata. Values are percentages calculated from the complete trajectories, not from plotting samples. The matrix exposes unsupported combinations, including the absence of mid-life and aged states for C27.

332 The 19 predeclared cases include the nominal baseline, two current-noise  
333 levels, voltage and temperature noise, three current-bias levels, voltage and  
334 temperature offsets, ADC quantization, initial SOC mismatch, periodic and  
335 random sample loss, three timing-jitter levels, a one-hour burst dropout, and  
336 randomized-sign voltage spikes. Gaussian sensor-noise cases use 10 determinis-  
337 tic seeds; random sample loss, jitter, and spike cases use 5 seeds; deterministic  
338 cases run once. Figure 10 maps each case to the manipulated measurement  
339 or state and marks the selected paired reference-SOH ablations. Input distur-  
340 bances are applied directly to measured channels, initialization errors only where  
341 structurally equivalent, and signal-integrity scenarios to sample availability or  
342 timing while preserving causal estimator execution.

### 343 3.4. Implementation details

344 Every estimator is executed causally, and all auxiliary quantities required by  
345 the models are rebuilt directly from disturbed inputs. This applies to  $Q_c$ , EFC,  
346  $dU/dt$ ,  $dI/dt$ , and hourly SOH aggregates, so no hidden dataset-side feature  
347 leaks into the estimator input. The SOH LSTM forwards hidden and cell states  
348 across hourly updates after its 192-h initialization context. The primary SOC-

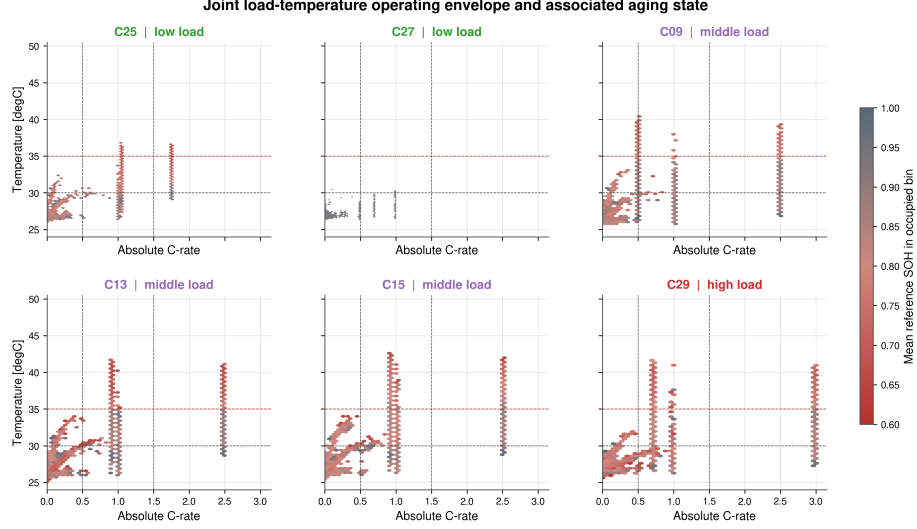


Figure 8: Joint absolute-C-rate and temperature operating envelopes of the six holdout cells. Color denotes the mean reference SOH in each occupied bin. Dashed lines show the fixed instantaneous-load and temperature-state boundaries used for stratified performance reporting.

GRU evaluation preserves its trained sequence-to-one semantics: each prediction uses the latest 2024-sample rolling window with recurrent state reset between windows. A faster continuous-stateful stream is treated only as a deployment ablation because it changed full-trajectory DD accuracy and therefore cannot silently replace the trained benchmark semantics. In freeze and dropout scenarios, disturbed or frozen measurements propagate consistently into derived online features. Direct-measurement estimators use the same CV-triggered reset that defines the full-charge start, whereas the DD initial-SOC test uses an equivalent offset of online  $Q_c$  because the network exposes no explicit SOC state. The environment enforces identical capacity assumptions, disturbed streams, targets, and metrics across all estimators.

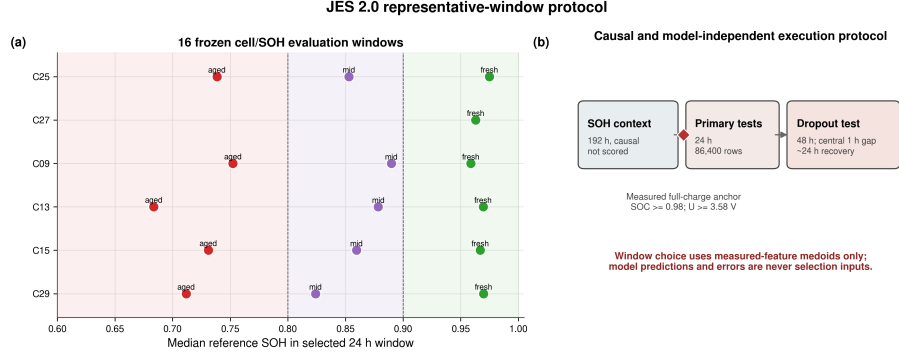


Figure 9: Frozen representative-window protocol. Panel (a) shows the 16 measured cell/SOH windows; absent states are not imputed. Panel (b) separates the unscored 192-h causal SOH context, the 24-h primary evaluation, and the 48-h dropout evaluation. Selection uses measured operating features only and is independent of estimator errors.

## 4. Results

### 4.1. Baseline performance

Figure 11 summarizes the nominal baseline performance on the selected benchmark trajectory. The hybrid direct-measurement model (HDM) achieves the lowest baseline error with  $\text{MAE} = 0.0024$ , followed by the hybrid equivalent-circuit model with 0.0090, the data-driven model with 0.0100, and the direct-measurement model with 0.0311, making HDM the clean-condition leader. This baseline ranking serves as the reference point for the following disturbed-measurement scenarios and for the corresponding robustness comparisons; the complete baseline metrics are listed in Appendix Table A.5.

### 4.2. Current-bias sensitivity

Figure 12 summarizes the adverse response to a signed current-gain sweep at  $\pm 0.5\%$ ,  $\pm 1.5\%$ , and  $\pm 3.0\%$ . For each magnitude and holdout cell, the larger MAE change from the two signs is reported because the opposite sign can partially compensate a pre-existing baseline error. The resulting worst-case degradation increases monotonically with gain-error magnitude for every estimator. DM and HDM show the strongest response, HECM is less sensitive, and DD is

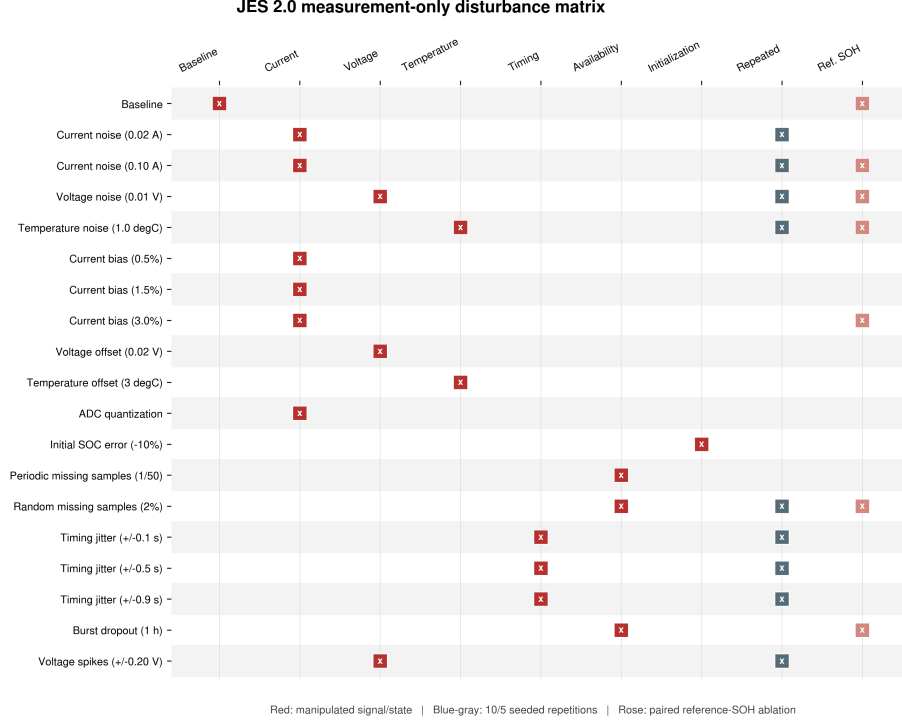


Figure 10: Complete JES2 measurement-only test matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel; blue-gray markers denote stochastic scenarios evaluated with the predeclared 10- or 5-seed budget across all available windows; rose markers identify selected paired reference-SOH ablations.

least sensitive. At 3%, the cell-macro MAE increases by 0.0107 for DM, 0.0100 for HDM, 0.0059 for HECM, and 0.0038 for DD.

#### 4.3. Noise sensitivity

Current noise is weaker than current bias in global MAE, but it remains informative through local behavior and output variability. In the compact matrix, the reported high-noise case uses  $\sigma_I = 0.10$  A, while the local detail sweep in Figure 13 extends the current-noise axis up to  $\sigma_I = 0.20$  A. Under the high-noise case, the hybrid equivalent-circuit model shows the largest global penalty ( $\Delta\text{MAE} = 0.0075$ ), whereas DM, HDM, and DD remain close to their baseline MAE values. The local noise story differs from the global metric: the data-

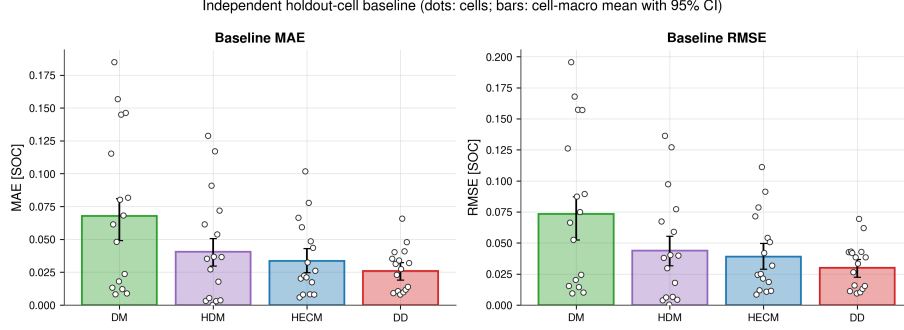


Figure 11: Baseline performance on the selected representative holdout-cell windows. The hybrid direct-measurement model provides the lowest clean-condition error, whereas the direct-measurement model shows the largest baseline deviation.

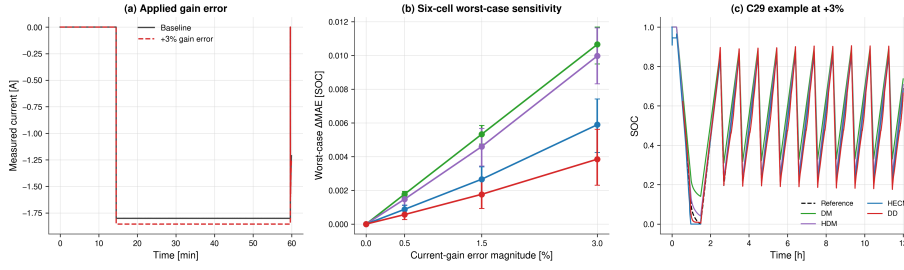


Figure 12: Current-gain sensitivity. (a) Representative measured-current segment before and after the +3% gain error. (b) Six-cell worst-case MAE change, retaining the adverse direction from each matched  $\pm b$  pair; error bars show 95% bootstrap confidence intervals across cells. (c) Representative C29 mid-life SOC trajectories at +3%.

387 driven model exhibits the strongest output variability because current noise  
 388 propagates into several derived input channels, whereas the hybrid equivalent-  
 389 circuit model is the most sensitive in the error domain. The broader noise  
 390 block is not limited to current noise: temperature noise with  $\sigma_T = 1.0^\circ\text{C}$  is  
 391 almost neutral for DM and HECM but increases the error of HDM and DD,  
 392 while voltage noise with  $\sigma_U = 0.01\text{V}$  is one of the strongest discriminators  
 393 because HECM remains nearly unaffected whereas DM, HDM, and DD de-  
 394 grade strongly. Figure 13 therefore separates the applied current disturbance, a  
 395 matched SOC comparison in the same window, the global MAE penalty, and the



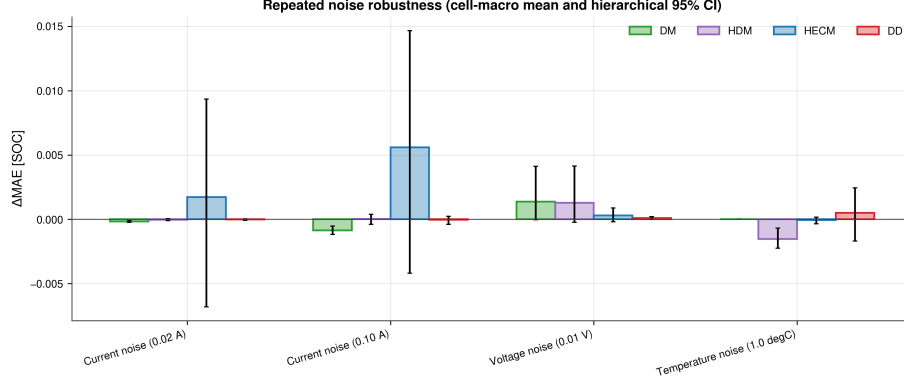


Figure 13: Noise sensitivity under additive current noise. The figure combines an applied current-noise example, a matched SOC comparison in the same 30 min window, the global MAE penalty, and the local output-variability sensitivity across noise levels.

local output-variability summary, while Appendix Tables A.6 and A.7 provide the complementary numerical evidence for current, voltage, and temperature noise.

#### 4.4. Initialization error and recovery

Initialization performance must be analyzed locally and in an evaluation window without constant-voltage reset events, because global MAE hides the actual re-synchronization behavior after a wrong initial SOC. The injected mismatch in the benchmark is  $-0.10$  SOC, as summarized in Table 3. DM recovers to the strict baseline band after about 1.17 h, HECM recovers after about 2.37 h under the strict criterion and after about 1.16 h under the fair criterion, whereas HDM does not re-enter either band in the inspected window. The data-driven model is evaluated through an initial perturbation of the online  $Q_c$  feature rather than through an explicit hard SOC-state reset, which keeps the test aligned with deployment-time observables; under this formulation it recovers after about 1.05 h under the strict criterion and after about 1.00 h under the fair criterion. Figure 14 visualizes the resulting reset-free recovery behavior, and the corresponding recovery metrics and baseline-band thresholds are listed in Appendix Table A.7.

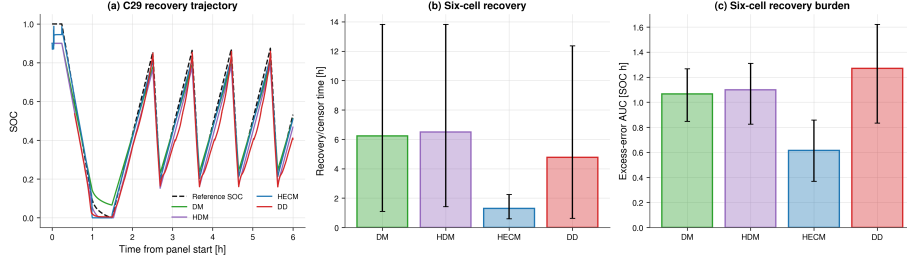


Figure 14: Initial-state recovery. (a) Representative C29 mid-life SOC trajectories during the first 6 h after the imposed mismatch. (b) Six-cell recovery or censor time. (c) Six-cell excess-error burden during recovery. Error bars show 95% bootstrap confidence intervals across cells.

#### 4.5. Missing samples and signal integrity

Missing samples are the strongest global signal-integrity stressor in the benchmark. In this scenario, every 50th sample is frozen, corresponding to one held sample every 50 s on the nominal 1 s grid; under this protocol the data-driven model changes only marginally from 0.0100 to 0.0117 MAE, whereas DM and HDM degrade much more strongly. The irregular-sampling scenario jitters the sampling intervals uniformly by  $\pm 0.1$  s around the nominal 1 s grid while preserving the full-run duration. Under this resampled implementation it remains nearly neutral for all models, with only very small signed MAE changes around zero. The burst-dropout scenario consists of one central frozen gap of 3600 s. It remains globally moderate, but all models require roughly 27–29 h to return to the baseline error band, making the post-gap recovery behavior more informative than the global MAE alone. Figure 15 summarizes the global signal-integrity penalties, while Figure 16 combines the local baseline-versus-dropout transition with the post-resume error and common 2% threshold. The corresponding global metrics are listed in Appendix Table A.6, the burst-dropout recovery times in Appendix Table A.7, and the scenario-to-evidence mapping in Appendix Table A.8.

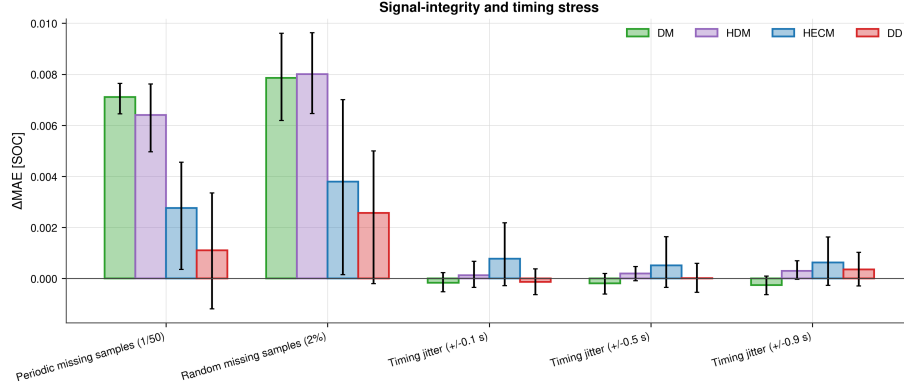


Figure 15: Signal-integrity performance summary. The left panel compares the global MAE penalties for missing samples, irregular sampling, and burst dropout, whereas the right panel summarizes the burst-dropout recovery time.

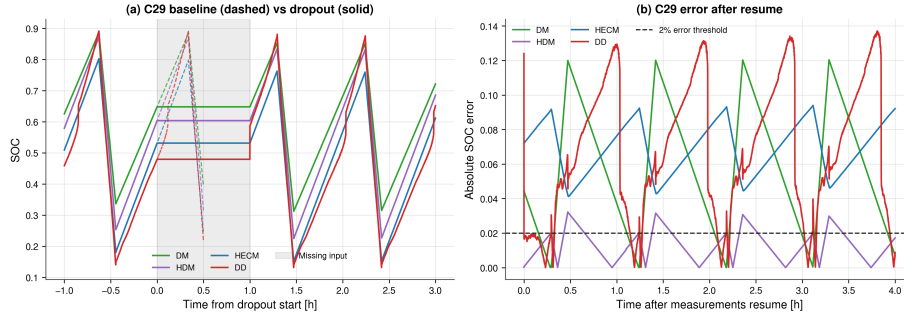


Figure 16: Transition through a 1 h burst dropout on the representative C29 mid-life window. (a) Baseline outputs (dashed) and dropout outputs (solid); the shaded region marks the missing-input interval. (b) Absolute SOC error after measurements resume; the dashed horizontal line marks the common 2% error threshold.

#### 4.6. Spike sensitivity

Single voltage spikes have different local effects across model classes even when their full-run MAE changes remain small. The protocol injects one  $\pm 0.20$  V voltage spike every 1000 samples, corresponding to every 1000 s on the nominal 1 s time base. At the correctly aligned representative C15 mid-life event, DD shows the strongest immediate deviation, while HECM exhibits a smaller rapidly decaying response. The six-cell direct-event analysis confirms this or-

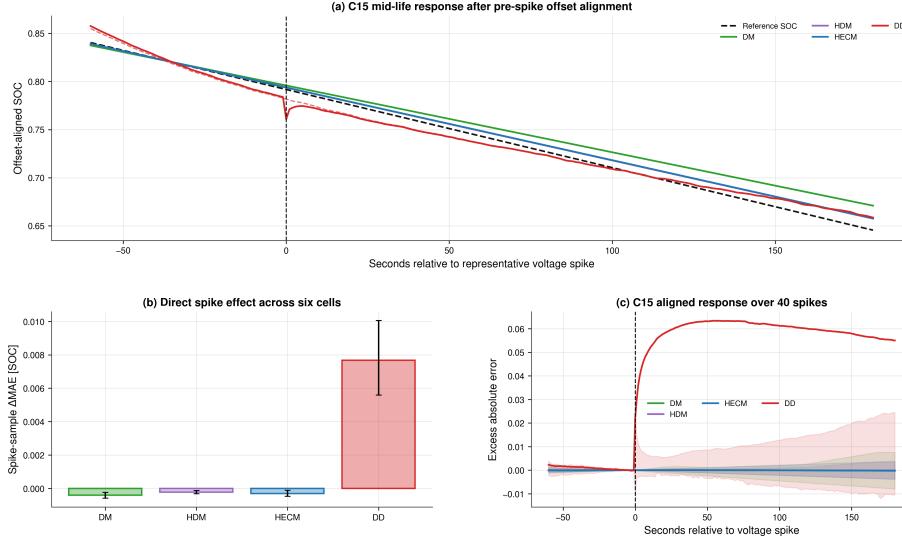


Figure 17: Voltage-spike response using current C15 and six-cell evidence. (a) Representative C15 mid-life trajectories aligned to the true injection sample and corrected only for each model’s constant mean pre-spike baseline offset; dashed model lines are paired baselines and solid lines are disturbed outputs. (b) Mean spike-sample  $\Delta$ MAE across six cells with 95% bootstrap confidence intervals. (c) Mean C15 excess absolute error aligned over 40 spikes; shaded bands show the interquartile range.

dering using the spike-sample MAE penalty relative to calm samples from the same run: DD has the largest mean penalty, including a C15 mid-life penalty of approximately  $4.8 \times 10^{-3}$  SOC compared with  $0.2 \times 10^{-3}$  SOC for HECM. Figure 17 reports the causal local transient; the numerical scenario-level metrics are summarized in Appendix Tables A.6 and A.7.

#### 4.7. Cross-scenario comparison

Figure 18 condenses the disturbed-scenario penalties into one cross-scenario view and makes the estimator-specific pattern shifts directly visible; it also includes the ADC-quantization case. The cross-scenario view confirms that the benchmark does not yield one universal winner across all criteria, because HDM is best under nominal conditions, HECM is most resilient under current bias and voltage noise, DD is the most stable estimator under missing samples, and

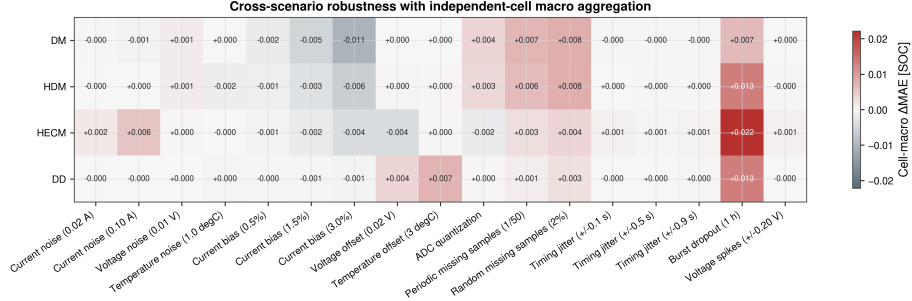


Figure 18: Cross-scenario error increase relative to the baseline case. The heatmap summarizes the disturbed-scenario  $\Delta\text{MAE}$  values, including ADC quantization, and highlights where robustness rankings change across disturbance classes.

DD also shows the fastest strict-band recovery after initialization mismatch. Current bias and voltage noise are the most discriminative scenarios in the present benchmark phase because they most clearly separate correction-capable estimators from purely integration-driven or fully data-driven alternatives. The heatmap is useful because it shows where the ranking changes across disturbance classes and where estimator strengths are scenario-specific rather than universal; the corresponding numerical overview is compiled in Appendix Tables A.5–A.7.

## 5. Discussion

### 5.1. Model-specific performance behavior

The direct-measurement model acts as the canonical drift-prone estimator, because every persistent current disturbance directly accumulates in the SOC estimate and missing measurement information cannot be corrected by an additional feedback mechanism. The hybrid direct-measurement model improves nominal calibration through online capacity adaptation, but it remains structurally exposed to the same current-integration failure mode and additionally inherits the dynamics of the learned SOH support. The hybrid equivalent-circuit model, combined here with the same shared data-driven online SOH estimator used by the other SOH-aware variants, benefits from interpretable ECM parameterization, explicit voltage-residual correction, and causal observer structure.

470 This combination explains why it handles current bias and voltage noise much  
471 better than the integration-based alternatives, even though it becomes more  
472 sensitive when noise perturbs the current-dependent parameter selection and  
473 observer update.

474 The present data-driven neural model behaves as a compact context-rich  
475 estimator whose robustness depends strongly on the disturbance channel: it  
476 is notably stable under missing samples, competitive under current bias, and  
477 the fastest model after initialization mismatch, but more exposed to derivative-  
478 amplified noise and voltage spikes. Read together, these results suggest that  
479 the current compact instance still relies too strongly on dominant trigger fea-  
480 tures from the input stream and does not yet exploit voltage and temperature  
481 information as effectively as the best hybrid physics-based alternative. That  
482 observation should be read as a design lesson rather than as a class limitation.  
483 Unlike the fixed-form direct-measurement and ECM estimators, the data-driven  
484 class retains a comparatively large tuning space through architecture, retraining,  
485 feature selection, and deployment-time state handling, so its present benchmark  
486 ranking reflects one compact embedded instance rather than a hard ceiling of  
487 the class itself.

## 488 5.2. Accuracy-performance trade-offs

489 The benchmark makes clear that low baseline MAE does not imply robust  
490 disturbed operation, because the best nominal model under clean measurements  
491 is not the best model under current bias or voltage noise. The observed trade-  
492 off is therefore not only between accuracy and computational complexity, but  
493 also between calibration strength, explicit physical correction capability, re-  
494 covery speed after state inconsistency, and sensitivity to disturbance propaga-  
495 tion through learned or derived features. The benchmark consequently yields  
496 a structured performance split: the hybrid direct-measurement model is the  
497 nominal-accuracy leader, the hybrid equivalent-circuit model is the overall ro-  
498 bustness leader, and the data-driven neural model converges back to the strict  
499 baseline band fastest after initialization mismatch. This result argues against

single-metric leaderboard thinking and favors a scenario-aware interpretation in which estimator quality is tied to the disturbance classes and recovery requirements that matter most in the target application.

### 5.3. Implications for BMS deployment

If current-sensor bias and long-term drift are dominant concerns, the benchmark evidence favors the hybrid equivalent-circuit approach, because the observer can re-anchor the SOC estimate through the measured voltage while the shared online SOH model keeps the parameterization aging-aware. If the deployment focus is nominal accuracy under controlled sensing and low-compute execution, the hybrid direct-measurement class remains attractive, but its vulnerability to biased current measurements must be considered explicitly. If rapid recovery after restart or initial-state mismatch is the primary requirement, the data-driven neural approach is attractive because it re-enters the strict baseline band fastest in the present benchmark, while simultaneously remaining strong under missing-sample stress. If robustness against signal-integrity issues such as missing samples is important, the data-driven model offers a strong alternative, provided that the input feature pipeline is designed so that the estimator does not over-focus on a single dominant measurement channel and remains stable under voltage-driven disturbances.

While the present study focuses on performance, practical BMS deployment also requires consideration of implementation complexity and scalability when the observed estimator behavior is transferred to real systems. Within that broader design space, the data-driven class retains the clearest improvement headroom because architecture, features, and training can be updated directly, whereas ECM-based estimators are more constrained by model topology and parameter-identification workflow. At the same time, the hybrid equivalent-circuit configuration used here shows why a physics-based SOC core plus shared data-driven SOH support can already yield a very strong compromise between interpretability, disturbance correction, and embedded feasibility. The footprint estimates indicate that persistent state memory is negligible for all estimators

in the present study and that all four deployment-prepared estimator variants remain below the reference 1 MB flash budget. The tightest flash case is the hybrid equivalent-circuit estimator because it combines the ECM lookup structure with the shared SOH-LSTM, whereas the direct-measurement baseline keeps the largest margin. For embedded BMS design, the practical selection criterion should therefore combine the expected fault profile of the measurement chain with implementation constraints. Table 4 condenses the benchmark outcomes into a decision-oriented recommendation map so that the reader is not left with a large set of disconnected scenario results.

Figure 19 complements this recommendation map with an illustrative visual synthesis of the same benchmark evidence. The values in this figure are relative scores rather than additional physical measurements. For each lower-is-better metric, the four estimator classes are min-max normalized as  $(x_{\max} - x)/(x_{\max} - x_{\min})$ , so that the best class in the present comparison receives 1 and the weakest class receives 0. The accuracy dimension averages the normalized baseline MAE, RMSE, and 95th-percentile error. The robustness dimension averages the normalized  $\Delta$ MAE values across the disturbed scenarios listed in Appendix Table A.8. The recovery dimension uses the strict recovery time after the initial-SOC perturbation; missing recovery within the inspected window is treated as non-recovery and assigned the lowest score. The bar chart then applies three illustrative priority weightings to these three dimensions: accuracy-weighted (0.60/0.20/0.20), robustness-weighted (0.20/0.60/0.20), and recovery-weighted (0.20/0.20/0.60), where the entries correspond to accuracy, robustness, and recovery, respectively. Thus, each profile assigns 60% of the composite score to its focal dimension and 20% to each of the other two dimensions. The accuracy-weighted profile should therefore be read as an accuracy-prioritized compromise, not as the pure baseline-accuracy ranking reported in Table 4. The synthesis is intended as a compact orientation aid and does not replace the raw scenario-wise results.



Table 4: Decision-oriented recommendation map derived from the benchmark results. The table summarizes which estimator class is most suitable when one deployment priority dominates the design decision.

| Primary deployment priority                                     | Recommended class | Main supporting benchmark evidence                                                                                                                                        | Main trade-off to consider                                                                                               |
|-----------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Lowest nominal SOC error under clean sensing                    | HDM               | Lowest baseline MAE on the benchmark trajectory (0.0024)                                                                                                                  | Strong sensitivity to current bias and no recovery to the baseline bands in the inspected initialization-mismatch window |
| Best overall robustness under uncertain measurement quality     | HECM              | Strongest current-bias robustness, nearly unchanged under voltage noise, and competitive baseline accuracy                                                                | Larger flash footprint and slower strict-band recovery after initialization mismatch than DD                             |
| Fastest re-synchronization after restart or wrong initial state | DD                | Fastest strict-band recovery in the local initialization analysis                                                                                                         | More sensitive than HECM to voltage spikes and less robust than HECM under current bias                                  |
| Highest resilience to missing samples and signal freezing       | DD                | Smallest MAE increase under missing samples and strong post-disturbance stability                                                                                         | Clean-condition accuracy remains below HDM, and voltage-driven disturbances require careful feature-pipeline design      |
| Best development headroom for iterative model improvement       | DD                | Architecture, features, and training can be updated directly while retaining embedded-feasible deployment and strong performance in recovery and missing-sample scenarios | Present compact instance is still weaker than HECM in the strongest bias- and voltage-driven stress cases                |
| Simplest implementation and largest resource margin             | DM                | Minimal architecture and smallest flash footprint by a wide margin                                                                                                        | Weakest robustness under biased current measurements and missing-sample stress                                           |

#### 5.4. Limitations and future extensions

The current benchmark phase is based on one representative dataset trajectory and should later be expanded to additional cells, broader aging states, and cross-cell validation before stronger generalization claims are made. The spike study currently uses isolated spike events; clustered disturbances, longer spike sequences, and hardware-level disturbance injection may reveal additional model differences. The initialization test for the data-driven model remains an online-state proxy based on  $Q_c$  perturbation rather than an explicit latent-state reset, which is methodologically justified but should still be interpreted as a class-specific approximation. Future extensions should include the remaining extended scenarios, additional cells, pack-level settings, and eventually

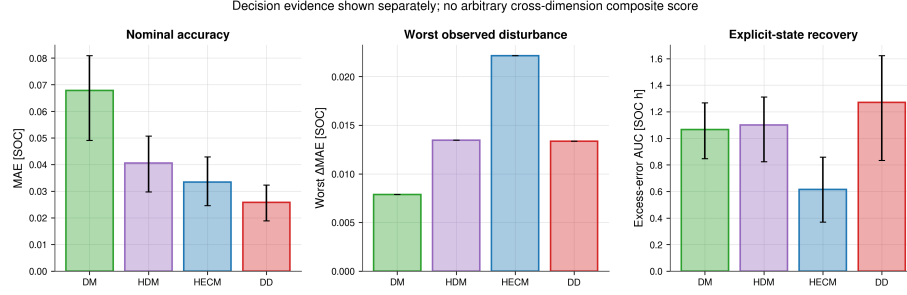


Figure 19: Decision-oriented synthesis of the benchmark results. Panel (a) collapses the benchmark into three relative, min-max-normalized decision dimensions, namely nominal accuracy, disturbed-scenario robustness, and recovery behavior. Panel (b) shows three illustrative deployment-priority profiles and is intended only as a visual complement to Table 4, not as a replacement for the raw scenario metrics.

570 hardware-in-the-loop or true embedded runtime benchmarks under the same  
 571 performance-oriented evaluation philosophy.

## 572 6. Conclusion

573 This work frames battery-state estimator evaluation as a performance prob-  
 574 lem under realistic measurement corruption rather than as a clean-data accuracy  
 575 comparison alone. The central engineering message is that estimator selection  
 576 depends on the expected disturbance profile of the BMS measurement chain, the  
 577 required recovery behavior after corrupted inputs, and the practical implemen-  
 578 tation constraints of the target system, because no single estimator dominates  
 579 all deployment-relevant dimensions. The main findings can be summarized by  
 580 estimator family: the hybrid direct-measurement model is the nominal-accuracy  
 581 leader, the hybrid equivalent-circuit model with shared data-driven SOH sup-  
 582 port is the strongest overall robustness choice in the most discriminative distur-  
 583 bance scenarios, and the data-driven neural model is strongest in recovery after  
 584 initialization mismatch and in missing-sample resilience.

585 If one general-purpose estimator must be preferred under the specific distur-  
 586 bance profile emphasized here, the benchmark supports the hybrid equivalent-

587 circuit approach as the strongest overall compromise. Its combination of in-  
 588 terpretable physical parameterization, voltage-based correction, and compact  
 589 shared SOH support preserves a clear role for ECM-based estimation in embed-  
 590 ded BMS design. At the same time, the data-driven model class remains highly  
 591 attractive when the development strategy includes iterative data collection and  
 592 model updates. The present benchmark also yields a practical design lesson:  
 593 data-driven SOC estimators should not rely too strongly on one dominant trig-  
 594 ger variable, but should be trained and feature-engineered to make balanced  
 595 use of the available current, voltage, temperature, and health context if they  
 596 are to realize their full robustness potential. The benchmark is designed to be  
 597 extensible, and the next steps are the extension to more cells and scenarios, the  
 598 inclusion of remaining disturbance classes, and the transfer of the methodology  
 599 toward broader embedded or hardware-in-the-loop validation.

## 600 **Appendix A. Supplementary Result Tables**

Table A.5: Baseline performance metrics for the selected bench-  
 mark trajectory.

| Class                     | MAE    | RMSE   | P95 error | Max error | Bias    |
|---------------------------|--------|--------|-----------|-----------|---------|
| Direct measurement        | 0.0311 | 0.0397 | 0.0800    | 1.0000    | 0.0311  |
| Hybrid direct measurement | 0.0024 | 0.0061 | 0.0094    | 1.0000    | 0.0022  |
| Hybrid ECM                | 0.0090 | 0.0127 | 0.0241    | 1.0000    | -0.0029 |
| Data-driven               | 0.0100 | 0.0143 | 0.0336    | 0.0686    | -0.0062 |

Table A.6: Cross-scenario global performance metrics used in the result discussion.

| Scenario             | Class | MAE    | $\Delta$ MAE | RMSE   | $\Delta$ RMSE | P95 error |
|----------------------|-------|--------|--------------|--------|---------------|-----------|
| Current noise (high) | DM    | 0.0310 | -0.0001      | 0.0394 | -0.0002       | 0.0795    |
| Current bias         | DM    | 0.3143 | 0.2833       | 0.3683 | 0.3286        | 0.6670    |
| Initial SOC error    | DM    | 0.0322 | 0.0011       | 0.0413 | 0.0017        | 0.0833    |
| Irregular sampling   | DM    | 0.0309 | -0.0001      | 0.0396 | -0.0001       | 0.0799    |
| Burst dropout        | DM    | 0.0332 | 0.0022       | 0.0487 | 0.0090        | 0.0819    |
| Missing samples      | DM    | 0.0386 | 0.0076       | 0.0476 | 0.0079        | 0.0926    |
| Voltage spikes       | DM    | 0.0312 | 0.0001       | 0.0398 | 0.0001        | 0.0801    |
| Temperature noise    | DM    | 0.0311 | 0.0000       | 0.0397 | 0.0000        | 0.0800    |
| Voltage noise        | DM    | 0.0524 | 0.0213       | 0.0648 | 0.0252        | 0.1228    |
| Current noise (high) | HDM   | 0.0044 | 0.0019       | 0.0074 | 0.0013        | 0.0120    |
| Current bias         | HDM   | 0.3068 | 0.3043       | 0.3638 | 0.3577        | 0.6668    |
| Initial SOC error    | HDM   | 0.0036 | 0.0011       | 0.0136 | 0.0075        | 0.0101    |
| Irregular sampling   | HDM   | 0.0027 | 0.0003       | 0.0061 | 0.0001        | 0.0095    |
| Burst dropout        | HDM   | 0.0051 | 0.0027       | 0.0325 | 0.0265        | 0.0098    |
| Missing samples      | HDM   | 0.0103 | 0.0079       | 0.0129 | 0.0069        | 0.0221    |
| Voltage spikes       | HDM   | 0.0062 | 0.0038       | 0.0140 | 0.0079        | 0.0328    |
| Temperature noise    | HDM   | 0.0067 | 0.0043       | 0.0110 | 0.0049        | 0.0210    |
| Voltage noise        | HDM   | 0.0751 | 0.0727       | 0.0908 | 0.0847        | 0.1578    |
| Current noise (high) | HECM  | 0.0165 | 0.0075       | 0.0280 | 0.0154        | 0.0701    |
| Current bias         | HECM  | 0.0932 | 0.0842       | 0.1303 | 0.1176        | 0.2954    |
| Initial SOC error    | HECM  | 0.0091 | 0.0001       | 0.0143 | 0.0016        | 0.0241    |
| Irregular sampling   | HECM  | 0.0095 | 0.0004       | 0.0131 | 0.0004        | 0.0248    |
| Burst dropout        | HECM  | 0.0104 | 0.0014       | 0.0228 | 0.0101        | 0.0246    |
| Missing samples      | HECM  | 0.0119 | 0.0029       | 0.0163 | 0.0036        | 0.0329    |
| Voltage spikes       | HECM  | 0.0090 | 0.0000       | 0.0127 | -0.0000       | 0.0241    |
| Temperature noise    | HECM  | 0.0090 | 0.0000       | 0.0127 | 0.0000        | 0.0241    |

| Scenario             | Class | MAE    | $\Delta$ MAE | RMSE   | $\Delta$ RMSE | P95 error |
|----------------------|-------|--------|--------------|--------|---------------|-----------|
| Voltage noise        | HECM  | 0.0090 | -0.0000      | 0.0127 | -0.0000       | 0.0241    |
| Current noise (high) | DD    | 0.0109 | 0.0009       | 0.0151 | 0.0008        | 0.0345    |
| Current bias         | DD    | 0.2859 | 0.2759       | 0.3426 | 0.3283        | 0.6367    |
| Initial SOC error    | DD    | 0.0110 | 0.0010       | 0.0175 | 0.0032        | 0.0355    |
| Irregular sampling   | DD    | 0.0105 | 0.0005       | 0.0144 | 0.0001        | 0.0330    |
| Burst dropout        | DD    | 0.0122 | 0.0022       | 0.0316 | 0.0173        | 0.0348    |
| Missing samples      | DD    | 0.0117 | 0.0016       | 0.0149 | 0.0006        | 0.0302    |
| Voltage spikes       | DD    | 0.0144 | 0.0044       | 0.0213 | 0.0070        | 0.0474    |
| Temperature noise    | DD    | 0.0181 | 0.0081       | 0.0223 | 0.0080        | 0.0451    |
| Voltage noise        | DD    | 0.0698 | 0.0597       | 0.0839 | 0.0696        | 0.1501    |

Table A.7: Local behavior and recovery metrics used in the result discussion.

| Class | Scenario          | Local metric                          | Value   | Threshold |
|-------|-------------------|---------------------------------------|---------|-----------|
| DM    | Initial SOC error | Recovery time to strict band [h]      | 1.1658  | 0.0800    |
| DM    | Initial SOC error | Recovery time to fair band [h]        | 0.2500  | 0.1599    |
| HDM   | Initial SOC error | Recovery time to strict band [h]      | –       | 0.0094    |
| HDM   | Initial SOC error | Recovery time to fair band [h]        | –       | 0.0200    |
| HECM  | Initial SOC error | Recovery time to strict band [h]      | 2.3693  | 0.0241    |
| HECM  | Initial SOC error | Recovery time to fair band [h]        | 1.1564  | 0.0482    |
| DD    | Initial SOC error | Recovery time to strict band [h]      | 1.0464  | 0.0336    |
| DD    | Initial SOC error | Recovery time to fair band [h]        | 1.0014  | 0.0672    |
| DM    | Burst dropout     | Recovery time after burst dropout [h] | 27.3511 | 0.0216    |
| HDM   | Burst dropout     | Recovery time after burst dropout [h] | 27.4093 | 0.0008    |
| HECM  | Burst dropout     | Recovery time after burst dropout [h] | 28.5611 | 0.0095    |
| DD    | Burst dropout     | Recovery time after burst dropout [h] | 27.4351 | 0.0070    |
| DM    | Voltage spikes    | Median spike recovery time [s]        | 0.0000  | 0.0800    |

| Class | Scenario             | Local metric                        | Value   | Threshold |
|-------|----------------------|-------------------------------------|---------|-----------|
| HDM   | Voltage spikes       | Median spike recovery time [s]      | 0.0000  | 0.0094    |
| HECM  | Voltage spikes       | Median spike recovery time [s]      | 0.0000  | 0.0241    |
| DD    | Voltage spikes       | Median spike recovery time [s]      | 0.0000  | 0.0336    |
| DM    | Current noise (high) | Late minus early excess rolling MAE | -0.0013 | –         |
| DM    | Current noise (high) | Mean excess rolling MAE             | -0.0001 | –         |
| HDM   | Current noise (high) | Late minus early excess rolling MAE | -0.0011 | –         |
| HDM   | Current noise (high) | Mean excess rolling MAE             | 0.0019  | –         |
| HECM  | Current noise (high) | Late minus early excess rolling MAE | 0.0038  | –         |
| HECM  | Current noise (high) | Mean excess rolling MAE             | 0.0073  | –         |
| DD    | Current noise (high) | Late minus early excess rolling MAE | -0.0002 | –         |
| DD    | Current noise (high) | Mean excess rolling MAE             | 0.0009  | –         |

Table A.8: Scenario overview linking disturbance configuration, primary evidence type, and main paper figure. This table clarifies which scenarios are interpreted predominantly through global metrics and which require additional local evidence.

| Scenario         | Configuration                                                                                      | Primary evidence                                 | Main figure(s) |
|------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------|
| ADC quantization | Steps of 0.01 A, 0.005 V, and 0.5 °C                                                               | Global $\Delta$ MAE; detailed values in Appendix | Figs. 18, B.20 |
| Current bias     | Constant current offset in the compact matrix, plus dedicated 0.5%/1.5%/3.0% sweep of $ I _{\max}$ | Global MAE degradation across the sweep          | Fig. 12        |

| Scenario           | Configuration                                                                         | Primary evidence                                                          | Main figure(s) |
|--------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------|
| Current noise      | Compact matrix with $\sigma_I = 0.10$ A; local detail sweep up to $\sigma_I = 0.20$ A | Global $\Delta$ MAE plus local output-variability / rolling-MAE evidence  | Fig. 13        |
| Voltage noise      | Additive Gaussian noise with $\sigma_U = 0.01$ V                                      | Global $\Delta$ MAE                                                       | Fig. 18        |
| Temperature noise  | Additive Gaussian noise with $\sigma_T = 1.0$ °C                                      | Global $\Delta$ MAE                                                       | Fig. 18        |
| Initial SOC error  | Additive initial mismatch of $-0.10$ SOC; DD via online $Q_c$ offset                  | Local recovery to strict and fair baseline bands; global MAE is secondary | Fig. 14        |
| Missing samples    | Every 50th sample frozen on the nominal 1 s grid                                      | Global $\Delta$ MAE                                                       | Figs. 15, 18   |
| Irregular sampling | Uniform sampling-time jitter of $\pm 0.1$ s around the nominal 1 s grid               | Global $\Delta$ MAE                                                       | Figs. 15, 18   |
| Burst dropout      | One central frozen gap of 3600 s                                                      | Global $\Delta$ MAE plus local transition and post-resume error           | Figs. 15, 16   |
| Voltage spikes     | One $\pm 0.20$ V single-sample spike every 1000 samples                               | Global $\Delta$ MAE plus aligned local excess-error response              | Figs. 17, 18   |

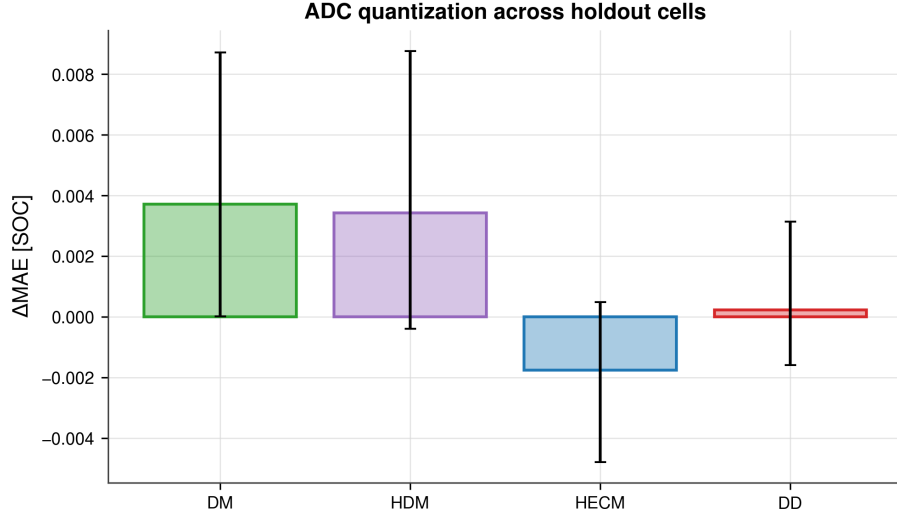


Figure B.20: ADC-quantization results. The two left panels illustrate representative current and voltage quantization in a local window, and the right panel summarizes the resulting global  $\Delta\text{MAE}$  relative to the baseline.

## Appendix B. ADC Quantization

ADC quantization is evaluated with step sizes of 0.01 A for current, 0.005 V for voltage, and 0.5 °C for temperature. Across all four estimator classes, the resulting deviations from the baseline remain small, so the detailed numerical comparison is collected here in the appendix while the scenario itself remains part of the overall benchmark.

Table B.9: ADC-quantization results for the four benchmarked estimator classes.

| Class | Baseline MAE | ADC MAE | $\Delta\text{MAE}$ |
|-------|--------------|---------|--------------------|
| DM    | 0.0311       | 0.0313  | +0.0002            |
| HDM   | 0.0024       | 0.0029  | +0.0005            |
| HECM  | 0.0090       | 0.0087  | -0.0003            |
| DD    | 0.0100       | 0.0098  | -0.0002            |



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612 The author contribution statement is provided in the non-blinded submission  
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614 **Declaration of competing interest**

615 The authors declare that they have no known competing financial interests or  
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617 in this paper.

618 **Declaration of generative AI and AI-assisted technologies in the manuscript  
619 preparation process**

620 During the preparation of this work the author(s) used ChatGPT, DeepL,  
621 and Grammarly in order to improve the linguistic quality and style of the  
622 manuscript. After using these tools, the author(s) reviewed and edited the con-  
623 tent as needed and take(s) full responsibility for the content of the published  
624 article.

625 **Data availability**

626 The underlying laboratory dataset is the MGFarm LFP 18650 cyclic-aging  
627 dataset described in [28]. The benchmark scripts and model runners are lo-  
628 cated in `DL\_Models/LFP\_SOC\_SOH\_Model/4\_simulation\_environment`,  
629 and the paper-facing figures and tables are collected under `DL\_Models/LFP\`

630 \_SOC\\_SOH\\_Model/4\\_simulation\\_environment/results. The data and code  
631 underlying this study will be made available on reasonable request.

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